

**AMENDMENT NO. 2 DECEMBER 1994**  
**TO**  
**IS 1606 :1979 SPECIFICATION FOR**  
**AUTOMOBILE LAMPS**  
**( Second Revision )**

( Page 6, clause 3.1 ) – Substitute the following for the existing clause :

**‘3.1** The various particulars of automobile lamps covered by the following seven groups are given in Tables 1 to 6:

- |           |                                                                                                                                                       |
|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| Group I   | Automobile lamps such as side, tail, dashboard and interior lighting lamps ( see Table 1 )                                                            |
| Group II  | Other types of interior lighting, stop-tail and direction indicator lamps which are bigger than Group I in size ( see Table 2 )                       |
| Group III | Lamps for headlights wherein the luminous intensity, focussing, etc, are important to give required beam pattern ( see Tables 3A, 3B, 3C, 3D and 3E ) |
| Group IV  | Festoon lamps ( see Table 4 )                                                                                                                         |
| Group V   | Lamps used for fog and revolving/flashing lights ( see Table 5 )                                                                                      |
| Group VI  | Wedge type bulbs (see Table 6)’                                                                                                                       |

( Page 7, clause 3.1.1 ) – Substitute the following for the existing clause:

**‘3.1.1 Typical Shapes of Lamps** – For easy identification, the typical shapes of various lamps are given in relevant tables.’

( Page 7, clause 4.1.1, line 3)– Delete the words ‘( see Tables 1 to 5 )’.

( Page 7, clause 4.1.4 ) – Substitute the following for the existing clause:

**‘4.1.4 Attachment of Caps to Glass Shells** – The caps shall be attached to the glass shells so strongly as to withstand torsion test carried out in accordance with 7.2.1 with the following torsional moments both during the inspection test and at the end of the life test:

| Types of Cap | Torsional Moment<br>Nm |
|--------------|------------------------|
| BA 7         | Under consideration    |
| BA 9         | 0.23                   |
| E 10         | 0.23                   |
| BA 15        | 1.15                   |
| BAY 15       | 1.15                   |

**Types of Cap** **Torsional Moment**

|              | Nm    |
|--------------|-------|
| P 36/21 x 22 | 1.15  |
| BA 20        | 1.15  |
| P 15d 25-1   | 1.15  |
| P 15d 25-3   | 1.15  |
| P 30d 10-3   | 1.15  |
| P 43t 38     | 1.15  |
| P 45t 41     | 1.15' |

( Page 8, Table 1 ) – Substitute the following for the existing table except for the figures :

**Table 1 Miniature Automobile Lamps ( Side, Tail and Dashboard Lamps )**  
( Clauses 3.1 and 4.1.1 )

| Ref No. | Volt | Watt | Diameter<br>Max | Overall<br>Length<br>Max | Pin to<br>Crown<br>Max | Length<br>Pin to<br>Solder<br>Max | Light<br>Centre<br>Length | Cap    | Ref to<br>Fig. | Initial Ratings of Lamps<br>(Average of RTQ) |      |      |       | Life Test  |                          | Lumen                        |                         |
|---------|------|------|-----------------|--------------------------|------------------------|-----------------------------------|---------------------------|--------|----------------|----------------------------------------------|------|------|-------|------------|--------------------------|------------------------------|-------------------------|
|         |      |      |                 |                          |                        |                                   |                           |        |                | Test Volts                                   | Watt |      | Lumen | Test Volts | Specified<br>Life<br>(h) | Minimum<br>Average<br>of LTQ | Mainten-<br>ance<br>(%) |
|         |      |      |                 |                          |                        |                                   |                           |        |                |                                              | Max  | Min  | Min   |            |                          |                              |                         |
| (1)     | (2)  | (3)  | (4)             | (5)                      | (6)                    | (7)                               | (8)                       | (9)    | (10)           | (11)                                         | (12) | (13) | (14)  | (15)       | (16)                     | (17)                         | (18)                    |
| M 6/1   | 6    | 0.6  | 6.8             | 20.7                     | 12                     | 8.7                               | –                         | BA 7S  | M <sub>1</sub> | 6.75                                         | 0.7  | 0.5  | –     | 6.75       | 200                      | 156                          | 85                      |
| M 6/2   | 6    | 1.2  | 6.8             | 20.7                     | 12                     | 8.7                               | –                         | BA 7S  | M <sub>1</sub> | 6.75                                         | 1.0  | 0.8  | –     | 6.75       | 200                      | 156                          | 85                      |
| M 6/3   | 6    | 2.0  | 8.8             | 23.9                     | 18.5                   | 5.9                               | –                         | BA 9S  | M <sub>2</sub> | 6.75                                         | 2.3  | 1.7  | 8     | 6.75       | 200                      | 156                          | 85                      |
| M 6/4   | 6    | 4.0  | 8.8             | 27.4                     | 21.5                   | 5.9                               | 15 ± 2                    | BA 9S  | M <sub>3</sub> | 6.75                                         | 4.4  | 3.6  | 26    | 6.75       | 200                      | 156                          | 85                      |
| M 6/5   | 6    | 5.0  | 19.0            | 37.5                     | 30                     | 7.5                               | 19 ± 2                    | BA 15S | M <sub>4</sub> | 6.75                                         | 5.5  | 4.5  | 37    | 6.75       | 200                      | 156                          | 85                      |
| M 6/6   | 6    | 10.0 | 19.0            | 37.5                     | 30                     | 7.5                               | 19 ± 2                    | BA 15S | M <sub>4</sub> | 6.75                                         | 11.0 | 9.0  | 100   | 6.75       | 200                      | 156                          | 85                      |
| M 6/7   | 6    | 3.0  | 8.8             | 23.9                     | 18.0                   | 5.9                               | 13 ± 2                    | BA 9S  | M <sub>2</sub> | 6.75                                         | 3.3  | 2.7  | 18    | 6.75       | 200                      | 156                          | 85                      |
| M 6/8   | 6    | 3.0  | 11.7            | 23.9                     | 18.0                   | 5.9                               | 12.7 ± 2                  | BA 9S  | M <sub>4</sub> | 6.75                                         | 3.3  | 2.7  | 18    | 6.75       | 200                      | 156                          | 85                      |
| M 12/1  | 12   | 1.2  | 6.8             | 20.7                     | 12                     | 8.7                               | –                         | BA 7S  | M <sub>1</sub> | 13.5                                         | 1.4  | 1.0  | –     | 13.5       | 200                      | 156                          | 85                      |
| M 12/2  | 12   | 2.0  | 6.8             | 20.7                     | 12                     | 8.7                               | –                         | BA 7S  | M <sub>1</sub> | 13.5                                         | 2.3  | 1.7  | 7     | 13.5       | 200                      | 156                          | 85                      |
| M 12/3  | 12   | 2.0  | 8.8             | 23.9                     | 18                     | 5.9                               | –                         | BA 9S  | M <sub>2</sub> | 13.5                                         | 2.3  | 1.7  | 7     | 13.5       | 200                      | 156                          | 85                      |
| M 12/4  | 12   | 4.0  | 8.8             | 27.4                     | 21.5                   | 5.9                               | 15 ± 2                    | BA 9S  | M <sub>2</sub> | 13.5                                         | 4.4  | 3.6  | 26    | 13.5       | 200                      | 156                          | 85                      |
| M 12/5  | 12   | 5.0  | 19.0            | 37.5                     | 30                     | 7.5                               | 19 ± 2                    | BA 15S | M <sub>4</sub> | 13.5                                         | 5.5  | 4.5  | 37    | 13.5       | 200                      | 156                          | 85                      |
| M 12/6  | 12   | 10.0 | 19.0            | 37.5                     | 30                     | 7.5                               | 19 ± 2                    | BA 15S | M <sub>4</sub> | 13.5                                         | 11.0 | 9.0  | 100   | 13.5       | 200                      | 156                          | 85                      |
| M 12/7  | 12   | 3.4  | 8.8             | 23.9                     | 18                     | 5.9                               | 13 ± 2                    | BA 9S  | M <sub>2</sub> | 13.5                                         | 3.8  | 3.0  | 17    | 13.5       | 200                      | 156                          | 85                      |
| M 12/8  | 12   | 3.4  | 11.7            | 23.9                     | 18.0                   | 5.9                               | 12.7 ± 2                  | BA 9S  | M <sub>4</sub> | 13.5                                         | 3.8  | 3.0  | 17    | 13.5       | 200                      | 156                          | 85                      |
| M 12/9  | 12   | 8.0  | 19.0            | 37.5                     | 30                     | 7.5                               | 19 ± 2                    | BA 15S | M <sub>4</sub> | 13.5                                         | 8.8  | 7.2  | 75    | 13.5       | 200                      | 156                          | 85                      |
| M 24/1  | 24   | 2.0  | 8.8             | 23.9                     | 18                     | 5.9                               | –                         | BA 9S  | M <sub>2</sub> | 28.0                                         | 3.5  | 2.5  | 12.0  | 28         | 200                      | 156                          | 85                      |
| M 24/2  | 24   | 4.0  | 8.8             | 27.4                     | 21.5                   | 5.9                               | 15 ± 2                    | BA 9S  | M <sub>3</sub> | 28.0                                         | 5.5  | 4.5  | 26.0  | 28         | 200                      | 156                          | 85                      |
| M 24/3  | 24   | 5.0  | 19.0            | 37.5                     | 30                     | 7.5                               | 19 ± 2                    | BA 15S | M <sub>4</sub> | 28.0                                         | 7.7  | 6.3  | 37.0  | 28         | 200                      | 156                          | 85                      |
| M 24/4  | 24   | 10.0 | 19.0            | 37.5                     | 30                     | 7.5                               | 19 ± 2                    | BA 15S | M <sub>4</sub> | 28.0                                         | 13.8 | 11.2 | 100   | 28         | 200                      | 156                          | 85                      |
| M 24/5  | 24   | 2.0  | 6.8             | 20.7                     | 12                     | 8.7                               | –                         | BA 7S  | M <sub>1</sub> | 28.0                                         | 3.5  | 2.5  | 12.0  | 28         | 200                      | 156                          | 85                      |

**NOTES**

1 Lamps with Ref No. M 6/1, M 6/2, M 6/7, M 12/7 and M 12/9 shall be phased out after 30 March 1995 and therefore are not intended for use in new designs.

2 For the dimension A, if agreed between the purchaser and the supplier, 7.2 and 9.5 are permissible in place of 6.8 and 8.8 respectively.

3 All dimensions are in millimetres.

4 Ageing period at test voltage shall be 15 minutes.

( Page 9, Table 2 ) – Substitute the following for the existing table except for the figures:

**Table 2 Medium Lamps ( Interior, Stop-and-Tail and Flasher Lamps )**  
( Clauses 3.1 and 4.1.1 )

| Ref No. | Volt | Watt | Diameter<br>Max | Overall<br>Length<br>Max | Pin to<br>Crown<br>Max | Length<br>Pin to<br>Solder<br>Max | Light<br>Centre<br>Length | Cap     | Ref to<br>Fig. | Initial Ratings of Lamps<br>(Average of RTQ) |      |      |       | Life Test  |                          | Lumen                        |                         |
|---------|------|------|-----------------|--------------------------|------------------------|-----------------------------------|---------------------------|---------|----------------|----------------------------------------------|------|------|-------|------------|--------------------------|------------------------------|-------------------------|
|         |      |      |                 |                          |                        |                                   |                           |         |                | Test Volts                                   | Watt |      | Lumen | Test Volts | Specified<br>Life<br>(h) | Minimum<br>Average<br>of LTQ | Mainten-<br>ance<br>(%) |
|         |      |      |                 |                          |                        |                                   |                           |         |                |                                              | Max  | Min  | Min   |            |                          |                              |                         |
|         |      |      | A               | B                        | C                      | D                                 | E                         |         |                |                                              |      |      |       |            |                          |                              |                         |
| (1)     | (2)  | (3)  | (4)             | (5)                      | (6)                    | (7)                               | (8)                       | (9)     | (10)           | (11)                                         | (12) | (13) | (14)  | (15)       | (16)                     | (17)                         | (18)                    |
| D 6/1   | 6    | 21   | 26.5            | 52.5                     | 45                     | 7 ± 0.5                           | 31.8 ± 2                  | BA 15s  | D <sub>1</sub> | 6.75                                         | 28   | 24   | 370   | 6.75       | 100                      | 85                           | 85                      |
| D 6/2   | 6    | 5    | 26.5            | 52.5                     | 45                     | 7 ± 0.5                           | 31.8 ± 2                  | BAY 15d | D <sub>2</sub> | 6.75                                         | 6.6  | 5.4  | 26    | 6.75       | 1000                     | 850                          | 85                      |
|         |      | 21   |                 |                          |                        |                                   |                           |         |                | 6.75                                         | 28   | 24   | 350   | 6.75       | 100                      | 85                           | 85                      |
| D 6/3   | 6    | 3    | 26.5            | 52.5                     | 45                     | 7 ± 0.5                           | 31.8 ± 2                  | BAY 15d | D <sub>2</sub> | 6.75                                         | 4.0  | 3.2  | 20    | 6.75       | 1000                     | 850                          | 85                      |
|         |      | 15   |                 |                          |                        |                                   |                           |         |                | 6.75                                         | 16.5 | 13.5 | 180   | 6.75       | 100                      | 85                           | 85                      |
| D 6/4   | 6    | 5    | 26.5            | 52.5                     | 45                     | 7 ± 0.5                           | 31.8 ± 2                  | BAY 15d | D <sub>2</sub> | 6.75                                         | 6.6  | 5.4  | 26    | 6.75       | 1 000                    | 850                          | 85                      |
|         |      | 10   |                 |                          |                        |                                   |                           |         |                | 6.75                                         | 11.0 | 9.0  | 90    | 6.75       | 100                      | 85                           | 85                      |
| D 6/5   | 6    | 5    | 26.5            | 52.5                     | 45                     | 7 ± 0.5                           | 31.8 ± 2                  | BAY 15d | D <sub>2</sub> | 6.75                                         | 6.6  | 5.4  | 26    | 6.75       | 1 000                    | 850                          | 85                      |
|         |      | 18   |                 |                          |                        |                                   |                           |         |                | 6.75                                         | 21.5 | 19.8 | 200   | 6.75       | 100                      | 85                           | 85                      |
| D 12/1  | 12   | 21   | 26.5            | 52.5                     | 45                     | 7 ± 0.5                           | 31.8 ± 2                  | BA 15s  | D <sub>1</sub> | 13.5                                         | 26.5 | 23.5 | 370   | 13.5       | 100                      | 85                           | 85                      |
| D 12/2  | 12   | 5    | 26.5            | 52.5                     | 45                     | 7 ± 0.5                           | 31.8 ± 2                  | BAY 15d | D <sub>2</sub> | 13.5                                         | 6.6  | 5.4  | 26    | 13.5       | 1 000                    | 850                          | 85                      |
|         |      | 21   |                 |                          |                        |                                   |                           |         |                | 13.5                                         | 26.5 | 23.5 | 350   | 13.5       | 100                      | 85                           | 85                      |
| D 12/3  | 12   | 5    | 26.5            | 52.5                     | 45                     | 7 ± 0.5                           | 31.8 ± 2                  | BAY 15d | D <sub>2</sub> | 13.5                                         | 6.6  | 5.4  | 26    | 13.5       | 1000                     | 850                          | 85                      |
|         |      | 10   |                 |                          |                        |                                   |                           |         |                | 13.5                                         | 11.0 | 9.0  | 90    | 13.5       | 100                      | 85                           | 85                      |
| D 12/4  | 12   | 10   | 26.5            | 52.5                     | 45                     | 7 ± 0.5                           | 31.8 ± 2                  | BA 15s  | D <sub>1</sub> | 13.5                                         | 11.0 | 9.0  | 100   | 13.5       | 100                      | 85                           | 85                      |
| D 12/5  | 12   | 15   | 26.5            | 52.5                     | 45                     | 7 ± 0.5                           | 31.8 ± 2                  | BA 15s  | D <sub>1</sub> | 13.5                                         | 16.5 | 13.5 | 180   | 13.5       | 100                      | 85                           | 85                      |
| D 12/6  | 12   | 10   | 26.5            | 52.5                     | 45                     | 7 ± 0.5                           | 31.8 ± 2                  | BA 15d  | D <sub>2</sub> | 13.5                                         | 11.0 | 9.0  | 90    | 13.5       | 1 000                    | 850                          | 85                      |
|         |      | 21   |                 |                          |                        |                                   |                           |         |                | 13.5                                         | 26.5 | 23.5 | 350   | 13.5       | 100                      | 85                           | 85                      |
| D 12/7  | 12   | 5    | 26.5            | 52.5                     | 45                     | 7 ± 0.5                           | 31.8 ± 2                  | BAY 15d | D <sub>2</sub> | 13.5                                         | 6.6  | 5.4  | 26    | 13.5       | 1000                     | 850                          | 85                      |
|         |      | 18   |                 |                          |                        |                                   |                           |         |                | 13.5                                         | 19.3 | 16.2 | 200   | 13.5       | 100                      | 85                           | 85                      |
| D 24/1  | 24   | 21   | 26.5            | 52.5                     | 45                     | 7 ± 0.5                           | 31.8 ± 2                  | BA 15s  | D <sub>1</sub> | 28                                           | 30   | 26   | 370   | 28         | 100                      | 85                           | 85                      |
| D 24/2  | 24   | 5    | 26.5            | 52.5                     | 45                     | 7 ± 0.5                           | 31.8 ± 2                  | BAY 15d | D <sub>2</sub> | 28                                           | 11.0 | 9.0  | 26    | 28         | 1 000                    | 850                          | 85                      |
|         |      | 21   |                 |                          |                        |                                   |                           |         |                | 28                                           | 30   | 26   | 350   | 28         | 100                      | 85                           | 85                      |
| D 24/3  | 24   | 15   | 26.5            | 52.5                     | 45                     | 7 ± 0.5                           | 31.8 ± 2                  | BA 15s  | D <sub>1</sub> | 28                                           | 16.5 | 13.5 | 180   | 28         | 100                      | 85                           | 85                      |
| D 24/4  | 24   | 21   | 26.5            | 52.5                     | 45                     | 7 ± 0.5                           | 31.8 ± 2                  | BA 15d  | D <sub>2</sub> | 28                                           | 30   | 26   | 370   | 28         | 100                      | 85                           | 85                      |

**NOTES**

- 1 Lamps D 6/2, D 6/4, D 12/5, D 12/7 shall be phased out and are not recommended for use in new design after 30 March 1995.
- 2 Ageing period at test voltage shall be 15 minutes.
- 3 All dimensions are in millimetres.

( Page 10, Table 3A ) – Substitute the following for the existing table except for the figures:

**Table 3A Head Light Lamps ( Anglo American Pattern )**

| Ref No. | Volt | Watt | Diameter<br>Max | Overall<br>Length<br>Max | Length<br>Pin to<br>Solder<br>Max | Light<br>Centre<br>Length | Cap | Filament | Initial Ratings of Lamps<br>(Average of RTQ) |      |      |       | Life Test |            | Lumen                    |                              |                         |
|---------|------|------|-----------------|--------------------------|-----------------------------------|---------------------------|-----|----------|----------------------------------------------|------|------|-------|-----------|------------|--------------------------|------------------------------|-------------------------|
|         |      |      |                 |                          |                                   |                           |     |          | Test Volts                                   | Watt |      | Lumen |           | Test Volts | Specified<br>Life<br>(h) | Minimum<br>Average<br>of LTQ | Mainten-<br>ance<br>(%) |
|         |      |      |                 |                          |                                   |                           |     |          |                                              | Max  | Min  | Min   | Max       |            |                          |                              |                         |
| (1)     | (2)  | (3)  | (4)             | (5)                      | (6)                               | (7)                       | (8) | (9)      | (10)                                         | (11) | (12) | (13)  | (14)      | (15)       | (16)                     | (17)                         | (18)                    |
| HA 6/1  | 6    | 30   | 29              | 64                       | 16 ± 2                            | 21.5 ± 0.5                | P36 | Driving  | 6.3                                          | 35   | 29   | 470   | —         | 7          | 50                       | 42                           | 85                      |
|         |      | 24   |                 |                          |                                   |                           |     | Passing  | 6.3                                          | 28   | 23   | 360   | 445       | 7          | 50                       | 42                           | 85                      |
| HA 12/1 | 12   | 50   | 29              | 64                       | 16 ± 2                            | 21.5 ± 0.5                | P36 | Driving  | 13.2                                         | 63   | 53   | 890   | —         | 14         | 75                       | 63                           | 85                      |
|         |      | 40   |                 |                          |                                   |                           |     | Passing  | 13.2                                         | 50   | 42   | 670   | 830       | 14         | 100                      | 85                           | 85                      |
| HA 24/1 | 24   | 54   | 29              | 64                       | 16 ± 2                            | 21.8 ± 0.5                | P36 | Driving  | 28.0                                         | 74   | 62   | 1 100 | —         | 28         | 125                      | 100                          | 85                      |
|         |      | 44   |                 |                          |                                   |                           |     | Passing  | 28.0                                         | 60   | 51   | 760   | 950       | 28         | 150                      | 120                          | 85                      |

**NOTES**

1 All dimensions are in millimetres.

2 Ageing period at test voltage shall be 15 minutes for HA 6/1 and 30 minutes for other lamps.

3 The lamps HA 6/1, HA 12/1, and HA 24/1 shall be phased out and are not recommended for new design after 30 March 1995.

(Page 11, Table 3B, figure ) – Delete values after 'f' from the figure.

( Page 11, Table 3B ) – Substitute the following for the existing table except for the figures:

**Table 3B Head Light Lamps ( Symmetric Beam Pattern )**

| Ref No. | Volt Watt |     | Diameter<br>Max | Overall Length<br>Max | Dimension<br>f | Length Pin to Solder<br>Max | Light Centre Length | Cap    | Filament | Initial Ratings of Lamps<br>(Average of RTQ) |      |      |       |      | Life Test  |                    | Lumen                  |                 |
|---------|-----------|-----|-----------------|-----------------------|----------------|-----------------------------|---------------------|--------|----------|----------------------------------------------|------|------|-------|------|------------|--------------------|------------------------|-----------------|
|         |           |     |                 |                       |                |                             |                     |        |          | Test Volts                                   | Watt |      | Lumen |      | Test Volts | Specified Life (h) | Minimum Average of LTQ | Maintenance (%) |
|         |           |     |                 |                       |                |                             |                     |        |          |                                              | Max  | Min  | Min   | Max  |            |                    |                        |                 |
| (1)     | (2)       | (3) | (4)             | (5)                   | (6)            | (7)                         | (8)                 | (9)    | (10)     | (11)                                         | (12) | (13) | (14)  | (15) | (16)       | (17)               | (18)                   | (19)            |
| HB 6/1  | 6         | 15  | 36              | 71                    | 1.8 ± 0.6      | 17.0                        | 32.7 ± 0.55         | BA 20d | Driving  | 6.75                                         | 16.5 | 13.0 | 180   | –    | 6.75       | 100                | 85                     | 85              |
|         |           | 15  |                 |                       |                |                             |                     |        | Passing  | 6.75                                         | 16.5 | 13.0 | 125   | 190  | 6.75       | 100                | 85                     | 85              |
| HB 6/2  | 6         | 25  | 36              | 71                    | 1.8 ± 0.6      | 17.0                        | 32.7 ± 0.55         | BA 20d | Driving  | 6.75                                         | 26.5 | 23.5 | 350   | –    | 6.75       | 100                | 85                     | 85              |
|         |           | 25  |                 |                       |                |                             |                     |        | Passing  | 6.75                                         | 26.5 | 23.5 | 250   | 375  | 6.75       | 100                | 85                     | 85              |
| HB 12/1 | 12        | 25  | 36              | 71                    | 1.8 ± 0.6      | 17.0                        | 32.7 ± 0.55         | BA 20d | Driving  | 13.5                                         | 26.5 | 23.5 | 350   | –    | 13.5       | 100                | 85                     | 85              |
|         |           | 25  |                 |                       |                |                             |                     |        | Passing  | 13.5                                         | 26.5 | 23.5 | 250   | 375  | 13.5       | 100                | 85                     | 85              |
| HB 12/2 | 12        | 35  | 36              | 71                    | 1.8 ± 0.6      | 17.0                        | 32.7 ± 0.55         | BA 20d | Driving  | 13.5                                         | 38.5 | 31.5 | 520   | –    | 13.5       | 100                | 85                     | 85              |
|         |           | 35  |                 |                       |                |                             |                     |        | Passing  | 13.5                                         | 38.5 | 31.5 | 370   | 560  | 13.5       | 100                | 85                     | 85              |
| HB 24/1 | 24        | 45  | 36              | 71                    | 2.2 ± 0.8      | 17.0                        | 32.7 ± 0.55         | BA 20d | Driving  | 28                                           | 49.5 | 40.5 | 665   | –    | 28         | 100                | 85                     | 85              |
|         |           | 40  |                 |                       |                |                             |                     |        | Passing  | 28                                           | 44.0 | 36.0 | 470   | 700  | 28         | 100                | 85                     | 85              |

#### NOTES

- 1 Lamp HB 6/1 shall be phased out and hence is not intended to be used in new design after 30 March 1995.
- 2 All dimensions are in millimetres.
- 3 Ageing period at test voltage shall be 15 minutes.

(Page 12, Table 3C, figure) – Delete the values after 'f' from the figure.

(Page 12, Table 3C, figure) – Substitute the following Table 3C for the existing table except for the figures:

**Table 3C Head Light Lamps (Asymmetric Beam Pattern)**  
(Clauses 3.1 and 4.1.1)

| Ref No. | Volt | Watt  | Dia-meter<br>Max | Overall Length<br>Max | Length Flange to Lug,<br>Max | Light Centre Length | Dimen-sion<br>f | Cap   | Filament | Initial Ratings of Lamps<br>(Average of RTQ) |      |      |       |      | Life Test  |                       | Lumen                  |                      |
|---------|------|-------|------------------|-----------------------|------------------------------|---------------------|-----------------|-------|----------|----------------------------------------------|------|------|-------|------|------------|-----------------------|------------------------|----------------------|
|         |      |       |                  |                       |                              |                     |                 |       |          | Test Volts                                   | Watt |      | Lumen |      | Test Volts | Specified Life<br>(h) | Minimum Average of LTQ | Mainten-<br>ance (%) |
|         |      |       |                  |                       |                              |                     |                 |       |          |                                              | Max  | Min  | Min   | Max  |            |                       |                        |                      |
| (1)     | (2)  | (3)   | (4)              | (5)                   | (6)                          | (7)                 | (8)             | (9)   | (10)     | (11)                                         | (12) | (13) | (14)  | (15) | (16)       | (17)                  | (18)                   | (19)                 |
| HC 6/1  | 6    | 45/40 | 41               | 82                    | 32                           | 28.5 ± 0.55         | 0.8 ± 0.6       | P 45t | Driving  | 6.0                                          | 49.5 | 40.5 | 600   | –    | 6.75       | 75                    | 63                     | 85                   |
|         |      |       |                  |                       |                              |                     |                 |       | Passing  | 6.0                                          | 44.0 | 36.0 | 400   | 550  | 6.75       | 150                   | 126                    | 85                   |
| HC 12/1 | 12   | 45/40 | 41               | 82                    | 32                           | 28.5 ± 0.55         | 1.8 ± 0.6       | P 45t | Driving  | 12                                           | 49.5 | 40.5 | 600   | –    | 13.5       | 75                    | 63                     | 85                   |
|         |      |       |                  |                       |                              |                     |                 |       | Passing  | 12                                           | 44.0 | 36.0 | 400   | 550  | 13.5       | 150                   | 126                    | 85                   |
| HC 24/1 | 24   | 55/50 | 41               | 82                    | 32                           | 28.5 ± 0.55         | 2.2 ± 0.8       | P 45t | Driving  | 24                                           | 60.5 | 49.5 | 600   | –    | 28.0       | 75                    | 63                     | 85                   |
|         |      |       |                  |                       |                              |                     |                 |       | Passing  | 24                                           | 55.0 | 45.0 | 400   | 550  | 28.0       | 150                   | 126                    | 85                   |

#### NOTES

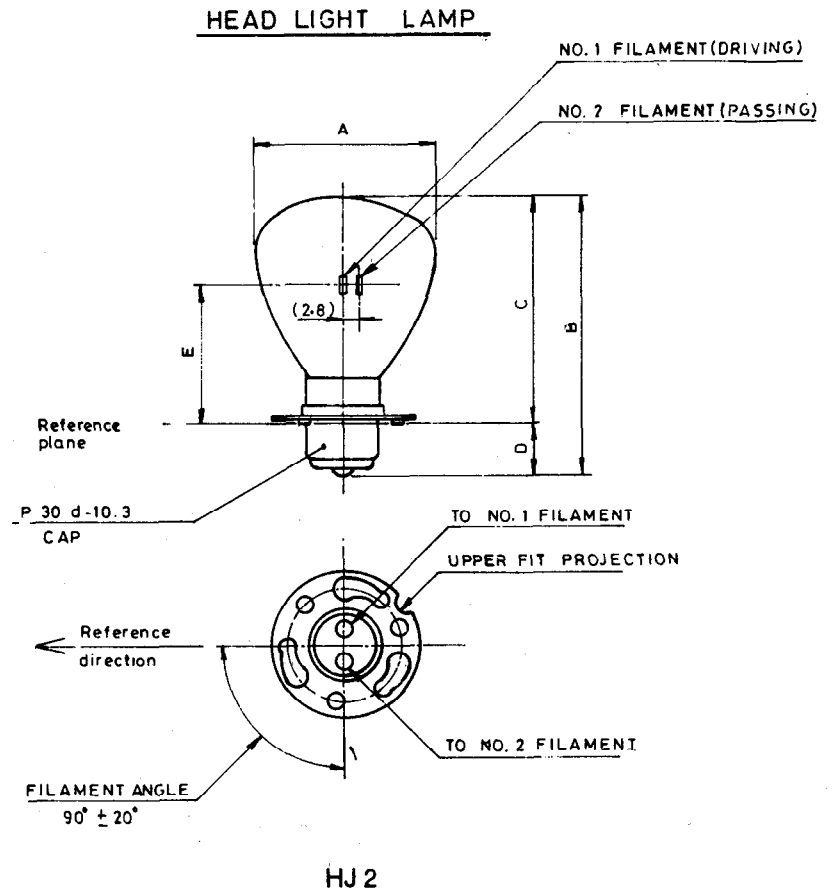
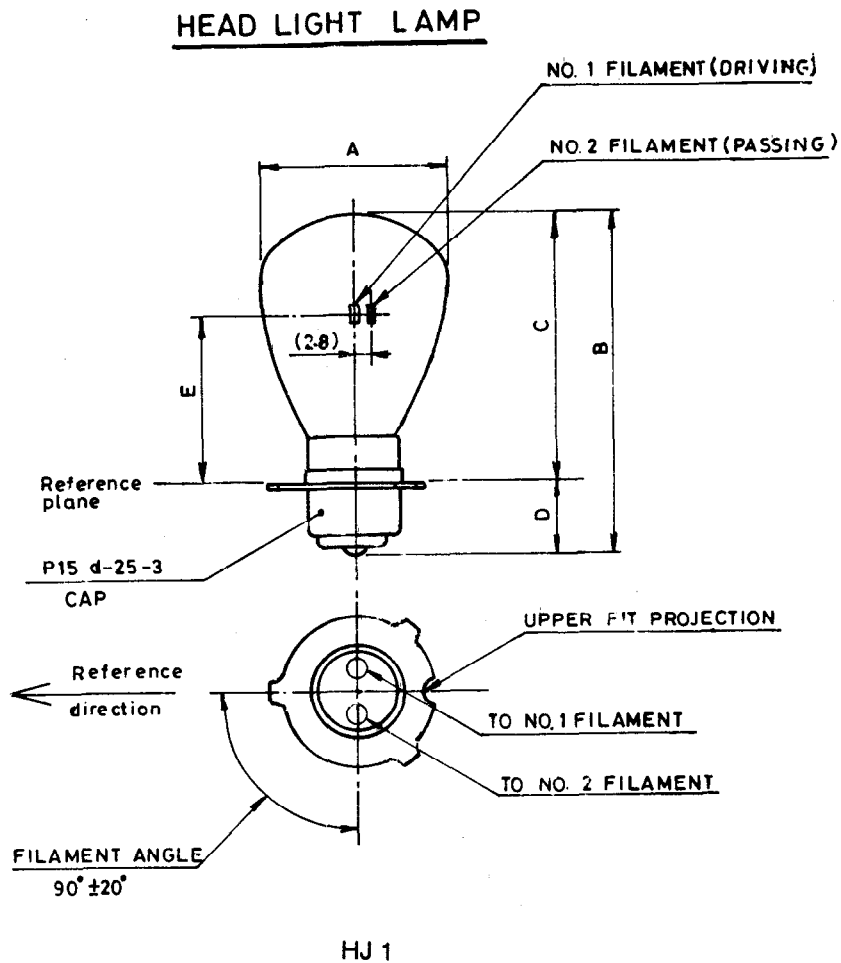
1 All dimensions are in millimetres.

2 Ageing period at test voltage shall be 30 minutes.

3 35 mm Oblong shape is also allowed.

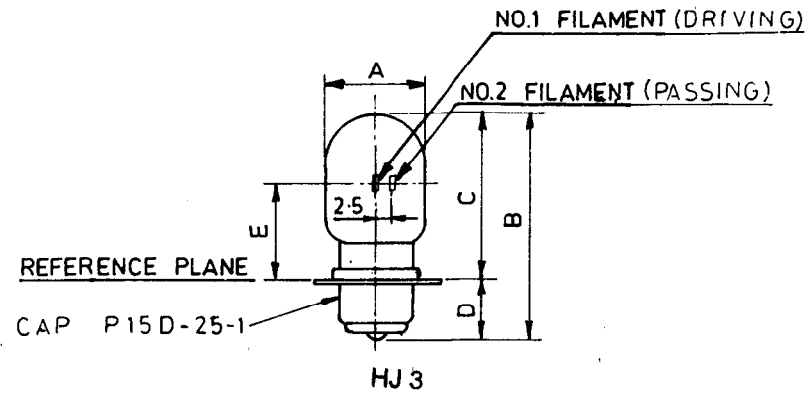
( Page 12, Table 3C ) – After Table 3C , insert the following new tables as Tables 3D and 3E :

**Table 3D Head Light Lamps ( Symmetric Beam Pattern )**



( Continued )

**Table 3D Head Light Lamps ( Symmetric Beam Pattern ) - Concluded**



| Ref No. | Volt | Watt | Dia-meter Max | Overall Length Max | Length Pin to Solder Max | Light Centre Length | Cap           | Ref to Fig. | Filament | Initial Ratings of Lamps (Average of RTQ) |      |      |       |      | Life Test  |                    | Lumen                  |                 |
|---------|------|------|---------------|--------------------|--------------------------|---------------------|---------------|-------------|----------|-------------------------------------------|------|------|-------|------|------------|--------------------|------------------------|-----------------|
|         |      |      |               |                    |                          |                     |               |             |          | Test Volts                                | Watt |      | Lumen |      | Test Volts | Specified Life (h) | Minimum Average of LTQ | Maintenance (%) |
|         |      |      |               |                    |                          |                     |               |             |          |                                           | Max  | Min  | Min   | Max  |            |                    |                        |                 |
| (1)     | (2)  | (3)  | (4)           | (5)                | (6)                      | (7)                 | (8)           | (9)         | (10)     | (11)                                      | (12) | (13) | (14)  | (15) | (16)       | (17)               | (18)                   | (19)            |
| HJ 6/1  | 6    | 25   | 31            | 60                 | 12                       | 28.5 ± 0.3          | P15d - 25 - 3 | HJ 1        | Driving  | 6.5                                       | 27.5 | 22.5 | 350   | -    | 6.50       | 100                | 85                     | 70              |
|         |      | 25   |               |                    |                          |                     |               |             | Passing  | 6.5                                       | 27.5 | 22.5 | 350   | 470  | 6.50       | 100                | 85                     | 70              |
| HJ 6/2  | 6    | 35   | 31            | 60                 | 12                       | 28.5 ± 0.3          | P15d - 25 - 3 | HJ 1        | Driving  | 6.5                                       | 38.5 | 31.5 | 570   | -    | 6.50       | 100                | 85                     | 70              |
|         |      | 35   |               |                    |                          |                     |               |             | Passing  | 6.5                                       | 38.5 | 31.5 | 570   | 720  | 6.50       | 100                | 85                     | 70              |
| HJ 6/3  | 6    | 25   | 36            | 60                 | 12                       | 28.5 ± 0.3          | P30d - 10.3   | HJ 2        | Driving  | 6.5                                       | 27.5 | 22.5 | 370   | -    | 6.50       | 100                | 85                     | 70              |
|         |      | 25   |               |                    |                          |                     |               |             | Passing  | 6.5                                       | 27.5 | 22.5 | 370   | 470  | 6.50       | 100                | 85                     | 70              |
| HJ 12/1 | 12   | 35   | 31            | 60                 | 12                       | 28.5 ± 0.3          | P15d - 25 - 3 | HJ 1        | Driving  | 13.0                                      | 38.5 | 31.5 | 580   | -    | 13.0       | 100                | 85                     | 70              |
|         |      | 35   |               |                    |                          |                     |               |             | Passing  | 13.0                                      | 38.5 | 31.5 | 580   | 750  | 13.0       | 100                | 85                     | 70              |
| HJ 12/2 | 12   | 25   | 20            | 50                 | 12                       | 20 ± 0.5            | P15d - 25 - 1 | HJ 3        | Driving  | 13.0                                      | 27.5 | 23.7 | 360   | -    | 13.0       | 100                | 85                     | 70              |
|         |      | 25   |               |                    |                          |                     |               |             | Passing  | 13.0                                      | 27.5 | 23.7 | 360   | 515  | 13.0       | 100                | 85                     | 70              |
| HJ 12/3 | 15   | 15   | 20            | 50                 | 12                       | 20 ± 0.5            | P15d - 25 - 1 | HJ 3        | Driving  | 13.0                                      | 16.5 | 13.5 | 200   | -    | 13.0       | 100                | 85                     | 70              |
|         |      | 15   |               |                    |                          |                     |               |             | Passing  | 13.0                                      | 16.5 | 13.5 | 200   | 260  | 13.0       | 100                | 85                     | 70              |

**NOTES**

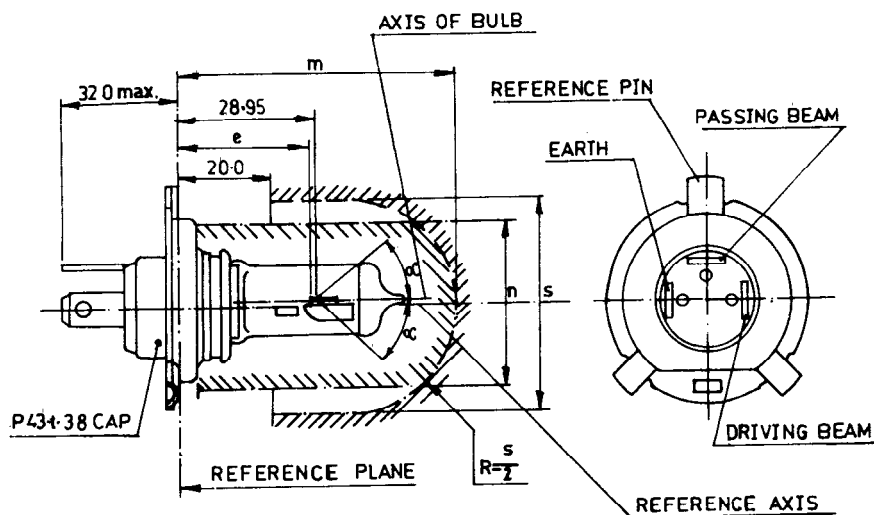
- 1 Lamps HJ 12/1 and HJ 12/3 shall be phased out and hence not intended for new design after 30 March 1995.
- 2 Ageing period at test voltage shall be 15 minutes.
- 3 All dimensions are in millimetres.
- 4 For the purpose of clause 4.3, the lumen maintenance (%) shall be verified at 75% of the specified life.

**Table 3E Head Light Lamps (Asymmetric Beam Pattern Halogen)**  
(Clauses 3.1 and 4.1.1)

| Ref No. | Volt | Watt  | Dia-meter<br>Max | Overall Length<br>Max | Length Pin to Solder<br>Max | Light Centre Length | Dimension<br>f | Cap       | Ref to Fig. | Fila-ment | Initial Ratings of Lamps<br>(Average of RTQ) |      |      |       |      | Life Test  |                    | Lumen                  |                 |
|---------|------|-------|------------------|-----------------------|-----------------------------|---------------------|----------------|-----------|-------------|-----------|----------------------------------------------|------|------|-------|------|------------|--------------------|------------------------|-----------------|
|         |      |       |                  |                       |                             |                     |                |           |             |           | Test Volts                                   | Watt |      | Lumen |      | Test Volts | Specified Life (h) | Minimum Average of LTQ | Maintenance (%) |
|         |      |       |                  |                       |                             |                     |                |           |             |           |                                              | Max  | Min  | Min   | Max  |            |                    |                        |                 |
| (1)     | (2)  | (3)   | (4)              | (5)                   | (6)                         | (7)                 | (8)            | (9)       | (10)        | (11)      | (12)                                         | (13) | (14) | (15)  | (16) | (17)       | (18)               | (19)                   | (20)            |
| HE 12/1 | 12   | 60/55 | -                | 92                    | 32                          | 28.5 + 0.55         | 1.7 + 0.5      | P43t - 38 | HE 1        | Driving   | 12                                           | 75   | -    | 1 000 | 1500 | 13.2       | 150                | 126                    | 85              |
|         |      |       |                  |                       |                             | - 0.45              | - 0.3          |           |             | Passing   | 12                                           | 68   | -    | 600   | 900  | 13.2       | 300                | 252                    | 85              |
| HE 12/2 | 12   | 60/55 | -                | 92                    | 32                          | 28.5 + 0.55         | 1.7 + 0.5      | P45t - 41 | HE 2        | Driving   | 12                                           | 75   | -    | 1000  | 1500 | 13.2       | 150                | 126                    | 85              |
|         |      |       |                  |                       |                             | - 0.45              | - 0.3          |           |             | Passing   | 12                                           | 68   | -    | 600   | 900  | 13.2       | 300                | 252                    | 85              |
| HE 24/1 | 24   | 75/70 | -                | 92                    | 32                          | 29.0 ± 0.60         | 2.0 ± 0.4      | P43t - 38 | HE 1        | Driving   | 24                                           | 85   | -    | 1000  | 1500 | 28.0       | 150                | 126                    | 85              |
|         |      |       |                  |                       |                             |                     |                |           |             | Passing   | 24                                           | 81   | -    | 640   | 960  | 28.0       | 300                | 252                    | 85              |
| HE 24/2 | 24   | 75/70 | -                | 92                    | 32                          | 29.0 ± 0.60         | 2.0 ± 0.4      | P45t - 41 | HE 2        | Driving   | 24                                           | 85   | -    | 1000  | 1500 | 28.0       | 150                | 126                    | 85              |
|         |      |       |                  |                       |                             |                     |                |           |             | Passing   | 24                                           | 81   | -    | 640   | 960  | 28.0       | 300                | 252                    | 85              |

**NOTES**

- 1 Halogen lamps shall be used only if the headlight assembly complies with IS 3563 : 1974 - Automobile Headlights.
- 2 Halogen lamps other than these shall be treated as non-standard.
- 3 HE 12/2 and HE 24/2 shall be phased out and are not intended for use in new design of head lamps after 30 March 1995.
- 4 All dimensions are in millimetres.
- 5 Ageing period at test voltage shall be 30 minutes.
- 6 For the purpose of clause 4.3, the lumen maintenance (%) shall be verified at 75% of the specified life.

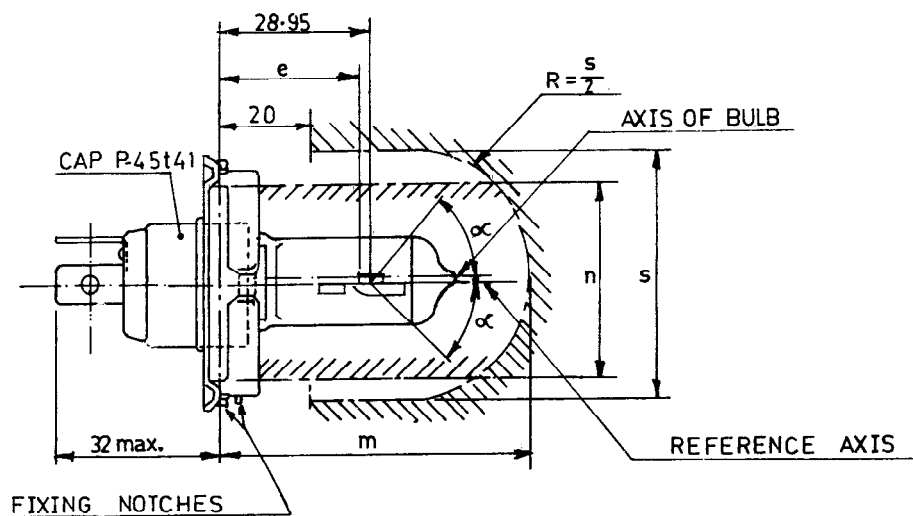


All dimensions in millimetres.

| Reference | Dimension | Tolerance        |
|-----------|-----------|------------------|
| <i>e</i>  | 28.5      | + 0.45<br>- 0.25 |
| <i>m</i>  | 60.0 Max  | —                |
| <i>n</i>  | 34.5 Max  | —                |
| <i>s</i>  | 45.0      | —                |
| $\alpha$  | 40° Max   | —                |

Halogen Lamp with P 43 t 38 Cap

HE 1



All dimensions in millimetres.

| Reference | Dimension | Tolerance        |
|-----------|-----------|------------------|
| <i>e</i>  | 28.5      | + 0.45<br>- 0.25 |
| <i>m</i>  | 60.0 Max  | —                |
| <i>n</i>  | 34.5 Max  | —                |
| <i>s</i>  | 45        | —                |
| $\alpha$  | 40° Max   | —                |

Halogen Lamp with P 45 t 41 Cap

HE 2

( Page 13, Table 4 ) – Substitute the following for the existing table except for the figures:

**Table 4 Festoon Lamps**

( Clauses 3.1 and 4.1.1 )

| Ref No. | Volt | Watt | Diameter<br>Max | Cap    | Ref to Fig.    | Initial Ratings of Lamps<br>(Average of RTQ) |      |      |       | Life Test  |                    | Lumen                  |                 |
|---------|------|------|-----------------|--------|----------------|----------------------------------------------|------|------|-------|------------|--------------------|------------------------|-----------------|
|         |      |      |                 |        |                | Test Volts                                   | Watt |      | Lumen | Test Volts | Specified Life (h) | Minimum Average of LTQ | Maintenance (%) |
|         |      |      |                 |        |                |                                              | Max  | Min  |       |            |                    |                        |                 |
| (1)     | (2)  | (3)  | (4)             | (5)    | (6)            | (7)                                          | (8)  | (9)  | (10)  | (11)       | (12)               | (13)                   | (14)            |
| F 6/1   | 6    | 3    | 9               | S 7    | F <sub>1</sub> | 6.75                                         | 3.9  | 3.2  | 22    | 6.75       | 200                | 156                    | 85              |
| F 6/2   | 6    | 5    | 11              | SV 8.5 | F <sub>2</sub> | 6.75                                         | 5.5  | 4.5  | 36    | 6.75       | 200                | 156                    | 85              |
| F 6/3   | 6    | 10   | 15.5            | SV 8.5 | F <sub>3</sub> | 6.75                                         | 11   | 9    | 100   | 6.75       | 200                | 156                    | 85              |
| F 6/4   | 6    | 18   | 15.5            | SV 8.5 | F <sub>3</sub> | 6.75                                         | 19.8 | 16.2 | 200   | 6.75       | 200                | 156                    | 85              |
| F 12/1  | 12   | 3    | 9               | S 7    | F <sub>1</sub> | 13.5                                         | 3.9  | 3.2  | 24    | 13.5       | 200                | 156                    | 85              |
| F 12/2  | 12   | 5    | 11              | SV 8.5 | F <sub>2</sub> | 13.5                                         | 5.5  | 4.5  | 36    | 13.5       | 200                | 156                    | 85              |
| F 12/3  | 12   | 10   | 15.5            | SV 8.5 | F <sub>3</sub> | 13.5                                         | 11   | 9    | 100   | 13.5       | 200                | 156                    | 85              |
| F 12/4  | 12   | 18   | 15.5            | SV 8.5 | F <sub>3</sub> | 13.5                                         | 19.8 | 16.2 | 200   | 13.5       | 200                | 156                    | 85              |
| F 12/5  | 12   | 5    | 9               | S 7    | F <sub>1</sub> | 13.5                                         | 5.5  | 4.5  | 36    | 13.5       | 200                | 156                    | 85              |
| F 12/6  | 12   | 10   | 11              | SV 8.5 | F <sub>2</sub> | 13.5                                         | 11   | 9    | 100   | 13.5       | 200                | 156                    | 85              |
| F 12/7  | 12   | 15   | 15.5            | SV 8.5 | F <sub>3</sub> | 13.5                                         | 16.5 | 13.0 | 150   | 13.5       | 200                | 156                    | 85              |
| F 24/1  | 24   | 5    | 11              | SV 8.5 | F <sub>2</sub> | 28                                           | 7.7  | 6.3  | 36    | 28         | 200                | 256                    | 85              |
| F 24/2  | 24   | 18   | 15.5            | SV 8.5 | F <sub>3</sub> | 28                                           | 19.8 | 16.2 | 200   | 28         | 200                | 156                    | 85              |

**NOTES**

- 1 Lamps F 12/1 shall be phased out and hence not intended for use in new design after 30 March 1995.
- 2 Maximum diameter of 12 mm is permissible in the place of 11 mm for lamps as for Fig. 2, if agreed by the supplier and the manufacturer.
- 3 All dimensions are in millimetres
- 4 Ageing period at test voltage shall be 15 minutes.

( Page 14, Table 5 ) – Substitute the following for the existing table:

**Table 5 Lamps Used for Fog and Revolving Flashing Lights**  
( Clauses 3.1 and 4.1.1 )

| Ref No. | Volt | Watt | Diameter<br>Max | Overall<br>Length<br>Max | Length<br>Pin to<br>Solder<br>Max | Light<br>Centre<br>Length | Cap    | Initial Ratings of Lamps<br>(Average of RTQ) |      |      |       | Life Test  |                          | Lumen                        |                         |
|---------|------|------|-----------------|--------------------------|-----------------------------------|---------------------------|--------|----------------------------------------------|------|------|-------|------------|--------------------------|------------------------------|-------------------------|
|         |      |      |                 |                          |                                   |                           |        | Test Volts                                   | Watt |      | Lumen | Test Volts | Specified<br>Life<br>(h) | Minimum<br>Average<br>of LTQ | Mainten-<br>ance<br>(%) |
|         |      |      |                 |                          |                                   |                           |        |                                              | Max  | Min  |       |            |                          |                              |                         |
|         |      |      |                 |                          |                                   |                           |        |                                              |      |      |       |            |                          |                              |                         |
| (1)     | (2)  | (3)  | (4)             | (5)                      | (6)                               | (7)                       | (8)    | (9)                                          | (10) | (11) | (12)  | (13)       | (14)                     | (15)                         | (16)                    |
| F 01    | 12   | 35   | 36              | 66.5                     | 16.5                              | 30.0 ± 0.55               | BA 20S | 13.5                                         | 38.5 | 31.5 | 685   | 11.5       | 100                      | 85                           | 85                      |
| F 02    | 12   | 50   | 29              | 64.0                     | 16.0                              | 21.5 ± 0.55               | P 36S  | 13.2                                         | 61.0 | 50.0 | 1150  | 14.0       | 75                       | 63                           | 85                      |
| F 03    | 24   | 36   | 36              | 58.2                     | 7.5                               | 31.8 ± 1.0                | BA 15S | 13.0                                         | 40.0 | 32.0 | 600   | 28.0       | 100                      | 85                           | 85                      |

**NOTES**

1 All dimensions are in millimetres.

2 Ageing period at test voltage shall be 15 minutes.

( Page 16, Table 7 ) – Delete.

( Page 17, Table 8A ) – Delete.

( Page 18, Table 8B ) – Delete.

( Page 19, Table 8C ) – Delete.

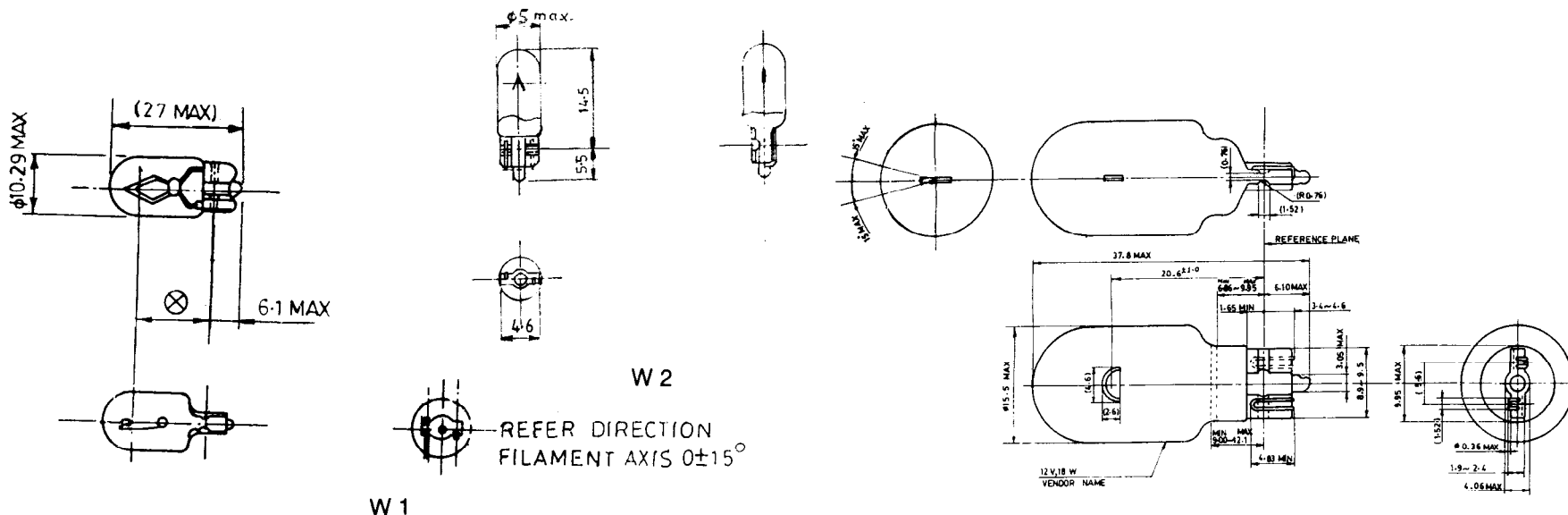
( Page 20, Table 9 ) – Delete.

( Page 20, Table 10 ) – Delete.

( Page 15, Table 6 ) – Substitute the following for the existing table:

**Table 6 Wedge Type Lamp**  
( Clauses 3.1, 4.2, 8.4 and 8.5 )

| Ref No. | Ratings |       | Ref to Fig. No. | Cap Size     | Agings Period at Life Test Volts | Initial Ratings of Lamps (Average of RTQ) |      |      |       | Life Test  |                    | Lumen                  |                 |
|---------|---------|-------|-----------------|--------------|----------------------------------|-------------------------------------------|------|------|-------|------------|--------------------|------------------------|-----------------|
|         | Volts   | Watts |                 |              |                                  | Test Volts                                | Watt |      | Lumen | Test Volts | Specified Life (h) | Minimum Average of LTQ | Maintenance (%) |
|         |         |       |                 |              |                                  |                                           | Max  | Min  |       |            |                    |                        |                 |
|         |         |       |                 |              |                                  |                                           |      |      |       |            |                    |                        |                 |
| (1)     | (2)     | (3)   | (4)             | (5)          | (6)                              | (7)                                       | (8)  | (9)  | (10)  | (11)       | (12)               | (13)                   | (14)            |
| W 12/1  | 12      | 1.2   | W 2             | W 2 x 4.6D   | 1/4                              | 13.5                                      | 1.4  | 1.0  | 6     | 13.5       | 1000               | 850                    | 85              |
| W 12/2  | 12      | 1.7   | W 1             | W 2.1 x 9.5D | 1/4                              | 14.4                                      | 2.0  | 1.5  | 8     | 14.8       | 1000               | 850                    | 85              |
| W 12/3  | 12      | 3.0   | W 1             | W 2.1 x 9.5D | 1/4                              | 13.5                                      | 3.4  | 2.6  | 15    | 13.5       | 1000               | 850                    | 85              |
| W 12/4  | 12      | 5     | W 1             | W 2.1 x 9.5D | 1/4                              | 13.5                                      | 5.5  | 4.5  | 37    | 13.5       | 200                | 156                    | 85              |
| W 12/5  | 12      | 18    | W 3             | W 2.1 x 9.5D | 1/4                              | 12.8                                      | 19.8 | 17.2 | 150   | 12.8       | 200                | 156                    | 85              |
| W 24/1  | 24      | 1.2   | W 2             | W 2 x 4.6D   | 1/4                              | 28                                        | 1.7  | 1.3  | 6     | 28         | 1000               | 850                    | 85              |
| W 24/2  | 24      | 3     | W 1             | W 2.1 x 9.5D | 1/4                              | 28                                        | 4.6  | 3.4  | 15    | 28         | 1000               | 850                    | 85              |
| W 24/3  | 24      | 5     | W 1             | W 2.1 x 9.5D | 1/4                              | 28                                        | 7.7  | 6.3  | 37    | 28         | 200                | 156                    | 85              |



\*Dimension 14.7 ± 2.3 for lamps W 12/3 and W 12/5 or 12.7 ± 1.5 for other lamps.

( Page 21, clause 4.2, line 3 ) Delete the words '(see Tables 6 to 10)'.

( Page 21, clause 4.3, lines 5 and 6 ) – Substitute the words '(see Table 6)' for '(see Tables 6 to 10)'.

( Page 22, clause 6.2 ) – Substitute the following for the existing clause:

**'6.2 The Inspection Test Quality (ITQ)** – Inspection test quality shall be as under:

i) In the case of double filament lamps given in Table 3E and single filament lamps: 30

ii) In the case of other double filament lamps: 60.

( Page 22, clause 6.6 ) – Delete.

( Page 23, clause 7.1.2 ) – Add the following at the end:

**'The lamps which fail during the ageing period shall be omitted from the test results.'**

( Page 24, clause 7.4, para 3 ) – Delete.

( Page 24, clause 7.4.1 ) – Substitute the following for the existing clause:

**'7.4.1 Voltage Control** – The measurements shall be carried out at the test voltage which shall be a stable ac voltage with frequency between 40 and 60 Hz. The test voltage is deemed to be stable when the momentary fluctuation does not exceed 1 percent and average over the period does not vary by more than + 0.1 V from the test voltage.'

( Page 24, clause 7.4.5 ) – Insert the following new clause as clause 7.4.6:

**'7.4.6 Test Conditions** – During the life test, it is preferable to maintain the room temperature at  $25 \pm 2.5^{\circ}\text{C}$ . If this is not possible, adequate care should be taken that heat of one lamp does not disturb the other lamps. This condition shall be deemed to be satisfied if a minimum distance of 250 mm is kept between lamps.'

[ Page 25, clause 8.4 (a) and (b) ] – Delete the words '(see Tables 6 to 10)' at both the places,

( Page 25, clause 3.5 ) – Renumber this clause as 8.5 and substitute the following for the existing clause:

**'8.5 Life Performance** – The batch shall be considered as complying with life performance requirements ( see 4.3 ), if:

a) The minimum average life of the LTQ shall not be less than the minimum life specified in the relevant tables.

NOTE – These figures are lower than the specified life in order to cover the statistical uncertainty involved in testing small samples.

b) The number of lamps of the LTQ having lives less than 60 percent of the specified life together with the lamps having lumen maintenance at 50 percent of the specified life less the value specified in the relevant table shall not exceed the compliance number given below:

| <i>Test Quantity</i> | <i>Compliance</i>  | <i>Number</i> |
|----------------------|--------------------|---------------|
| 10                   | For Group I and IV | 4             |
|                      | For Other Groups   | 2             |

( *Page 25, clause A-1*) – Insert the following new clause as clause **A-2** :

**‘A-2** In the case of caps BA 20d, BA 15, BAY 15, BA 9 and BA 7, the slot if provided for side solder of wire and soldering shall be such that the tolerance on diameter ‘A’ is maintained over the area covered by the dimensions ‘N’.

(*Page 27, Article Sheet No. 2*) – Add the following Note at the end:

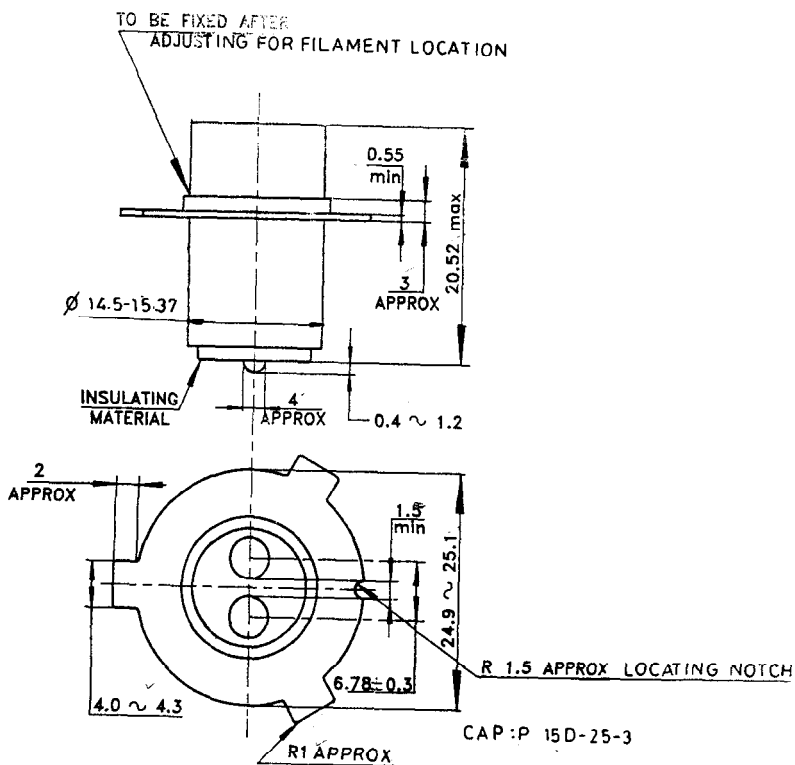
‘NOTE – The maximum dimensions *P* and *D* for Caps on finished lamps can be permitted to be 11.2 and 6.2 respectively if agreed to between purchaser and supplier.’

( *Page 31, Article Sheet No. 5, figure* ) – Substitute ‘K1 for oval contacts K2 for round contacts’ for the notation ‘K’.

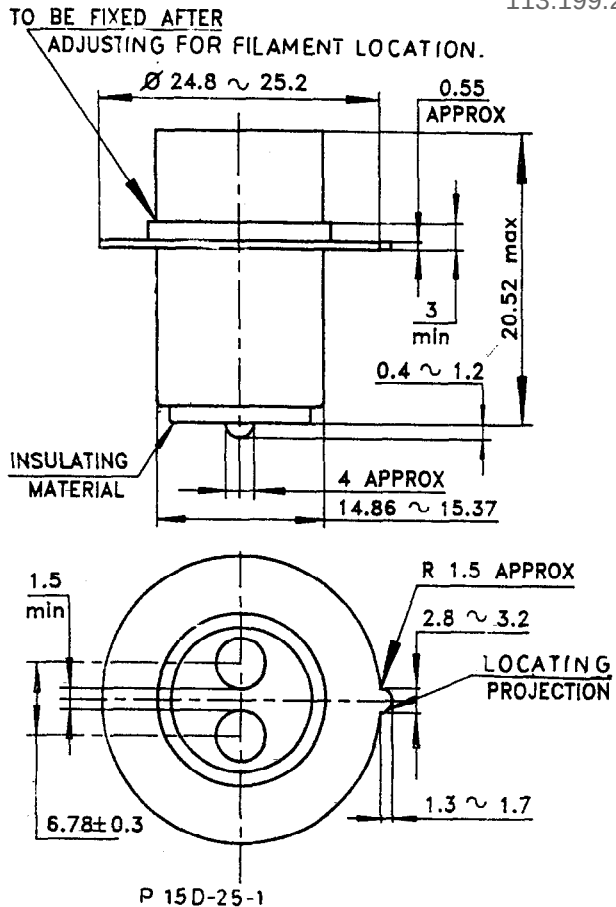
( *Page 31, Article Sheet No. 5, Figure* ) – Substitute ‘J1 for oval contacts J2 for round contacts’ for the notation ‘J’.

( *Page 34, Article Sheet No. 7, Internal Table* ) – For dimension ‘T’ substitute ‘0.65’ for ‘0.74’.

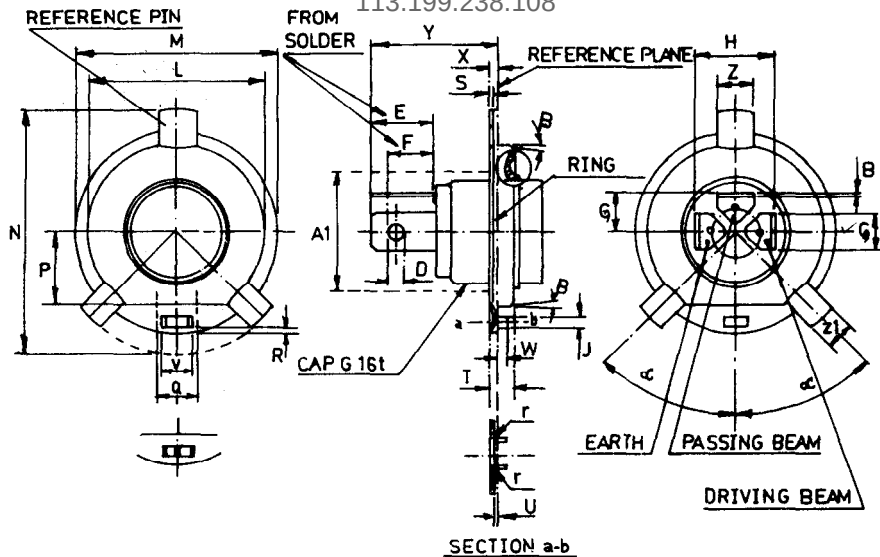
( *Page 34, Article Sheet No. 7* ) – Insert the following Article Sheets Nos. 8, 9, 10 and 11.



ARTICLE SHEET NO. 8



ARTICLE SHEET NO. 9



SECTION a-b

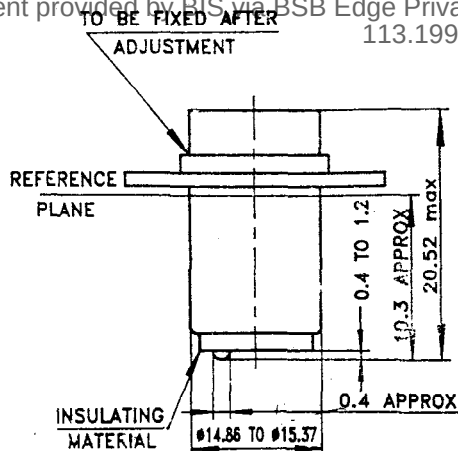
| Dimension | Min. | Max. | Dimension | Min. | Max. |
|-----------|------|------|-----------|------|------|
| A1        | 25.0 | —    | S         | 0.5  | —    |
| B         | 0.7  | 0.8  | T         | 5.0  | 6.0  |
| C         | 7.7  | 8.1  | U         | —    | *    |
| D         | 3.0  | 33   | V         | 6.3  | 6.5  |
| E         | 11.8 | 13.6 | W         | 1.8  | 2.2  |
| F         | 8.8  | 10.3 | X         | 1.1  | 1.3  |
| G         | 8.5  | 9.0  | Y         |      | 32.0 |
| H         | 17.0 | 17.9 | Z         | 7.9  | 8.0  |
| J         | 1.9  | 2.1  | Z1        | 5.8  | 6.2  |
| L         | 37.8 | 38.0 | r         | —    | *    |
| M         | 42.8 | 43.0 | $\alpha$  | 44°  | 46°  |
| N         | 51.6 | 52.0 | $\beta$   | —    | 5°   |
| P         | 15.3 | 15.5 |           |      |      |
| Q         | 8.5  | —    |           |      |      |
| R         | 1.3  | 1.7  |           |      |      |

\* Radius  $r$  must not exceed dimension  $U$ .

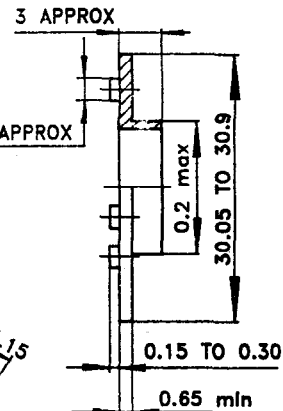
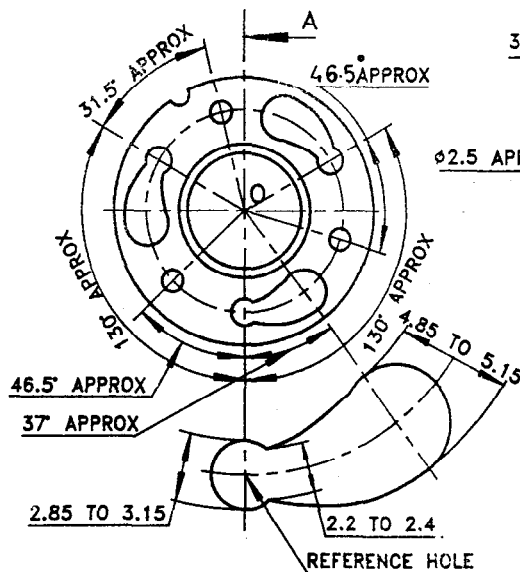
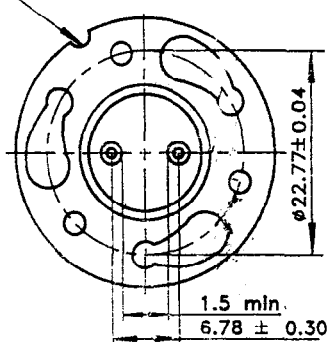
Cap P43t-38

ARTICLE SHEET NO. 10

( FTD 23 )



NOTCH TO SHOW UPPER  
DIRECTION R 2.4 APPROX



SECTION A-A

CAP: P 300-10-3

ARTICLE SHEET NO. 11